Customer Retention Analysis

AlphaCom Telecommunications — Customer Churn Prediction Project

Business Context

AlphaCom, a leading telecommunications provider, has recently experienced a concerning rise in customer churn despite offering competitive services and a wide product portfolio. This project analyzes customer demographic, account, and service usage data to identify the key drivers of churn and to build a predictive model that flags customers at high risk of leaving, so that AlphaCom can take proactive retention action.

Data & Methodology

The dataset contains 12,055 customer records across 20 features covering demographics (gender, senior citizen status, partner/dependents), account information (tenure, contract type, payment method, charges), and subscribed services (phone, internet, streaming, security add-ons). Data cleaning addressed invalid tenure values, currency-formatted charge fields, and inconsistent text casing/spacing in categorical fields (e.g. PaymentMethod, Churn). The cleaned data was split 70/15/15 into training, validation, and test sets, stratified on churn.

Key EDA Findings

- Customers on month-to-month contracts churn at a substantially higher rate than those on one- or two-year contracts.
- Customers with short tenure (under ~12 months) are far more likely to churn than long-tenured customers.
- Fiber optic internet subscribers show a higher churn rate than DSL or no-internet customers.
- Customers without Online Security or Tech Support add-ons churn more frequently.
- Electronic check as a payment method is associated with a higher churn rate than automatic payment methods.
- Churned customers tend to have higher average monthly charges but lower total charges (consistent with shorter tenure).

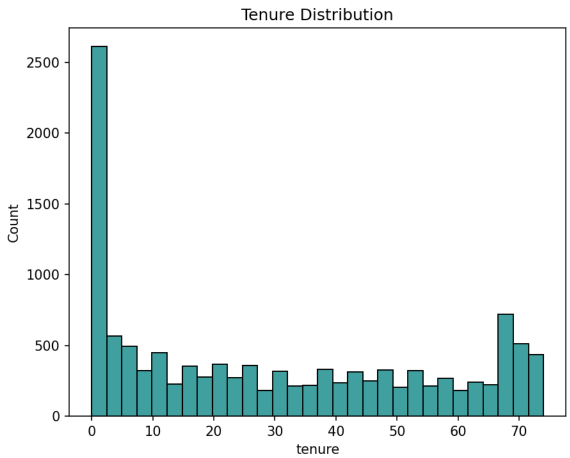
Modeling

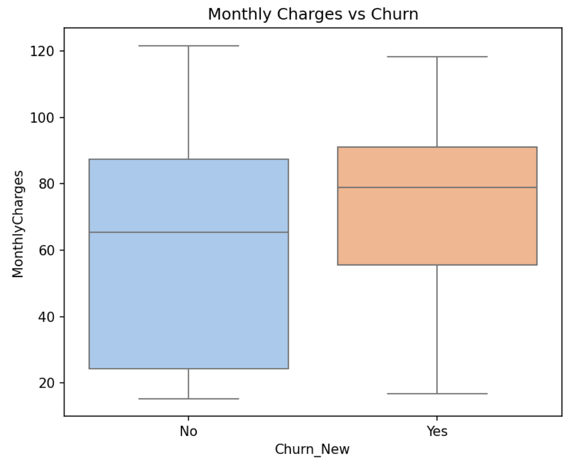
Multiple classification models were trained and evaluated, including Logistic Regression, Decision Tree, Bagging, Random Forest, AdaBoost, and Gradient Boosting. Hyperparameter tuning was performed on Random Forest, AdaBoost, and Gradient Boosting. The tuned Gradient Boosting model delivered the strongest and most balanced performance (recall, precision, and ROC-AUC) across training, validation, and held-out test data, and was selected as the final model.

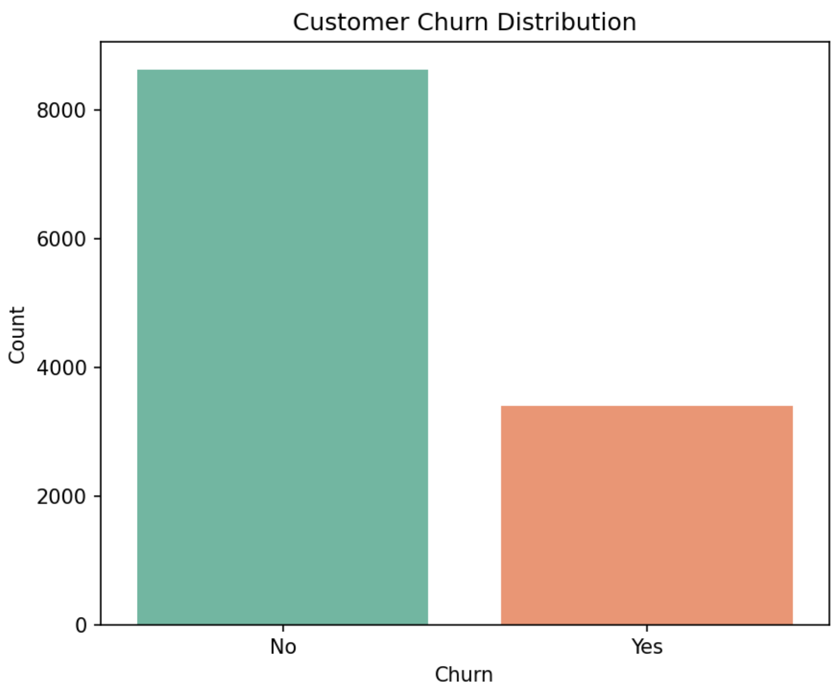
Recommendations

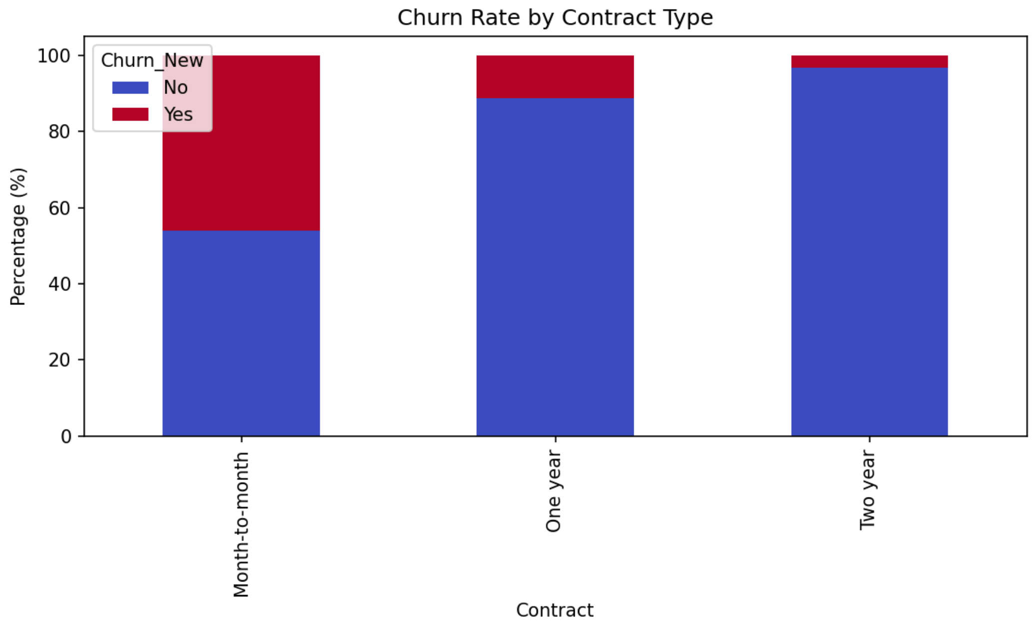
- Prioritize retention outreach for new, month-to-month customers within their first 12 months.
- Bundle Online Security and Tech Support with Fiber optic plans to reduce churn risk.
- Encourage migration from Electronic Check to automatic payment methods.
- Offer incentives for upgrading from month-to-month to annual contracts.
- Deploy the churn model to score the active customer base monthly and trigger proactive retention campaigns.

Supporting Visuals









Churn Rate by Contract Type
Churn_New
No
Yes
Percentage (%)
0
20
40
60
80
100
Month-to-month
One year
Two year
Contract